Blind Source Separation

Václav Šmídl

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Recapitulation

1. Model-based development:
   - modelling assumptions $\Longrightarrow$ estimation method
   - Gaussian Mixture $\Longrightarrow$ GMM method (via the EM algorithm)
   - Linear combination with Gaussian noise $\Longrightarrow$ OLS
   - mixture of linear models $\Longrightarrow$ custom algorithm

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   - mixture of linear models $\Longrightarrow$ custom algorithm
2. Method-based development
   - PCA $\Longrightarrow$ OLS $\Longrightarrow$ GMM

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Consider two least squares problem

$$y_1 = X\theta_1 \qquad\qquad y_2 = X\theta_2 \qquad\qquad Y = X\Theta$$

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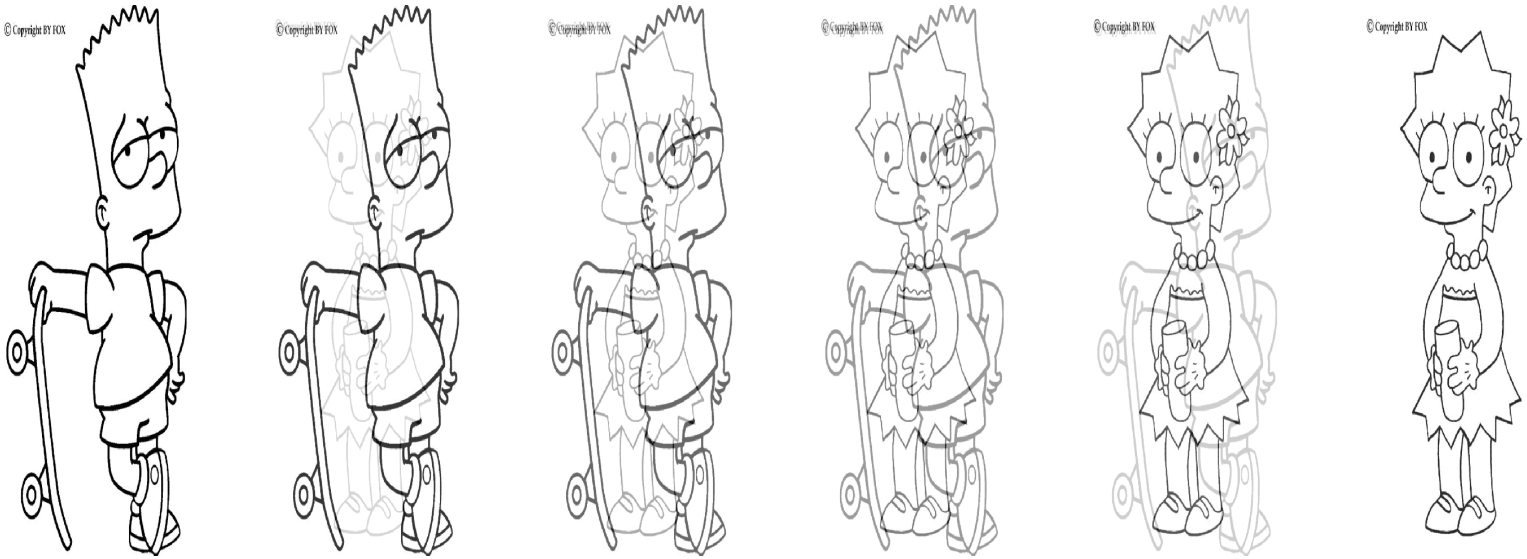
In probabilities, matrix Normal distribution

$$\begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \hat{\theta}_1 \\ \hat{\theta}_2 \end{bmatrix}, \begin{bmatrix} (X^T X)^{-1} & 0 \\ 0 & (X^T X)^{-1} \end{bmatrix} \right) <=> \Theta = \mathcal{N}\left(\hat{\Theta}, I \otimes (X^T X)^{-1} \right)$$

Blind source separation – image sequence



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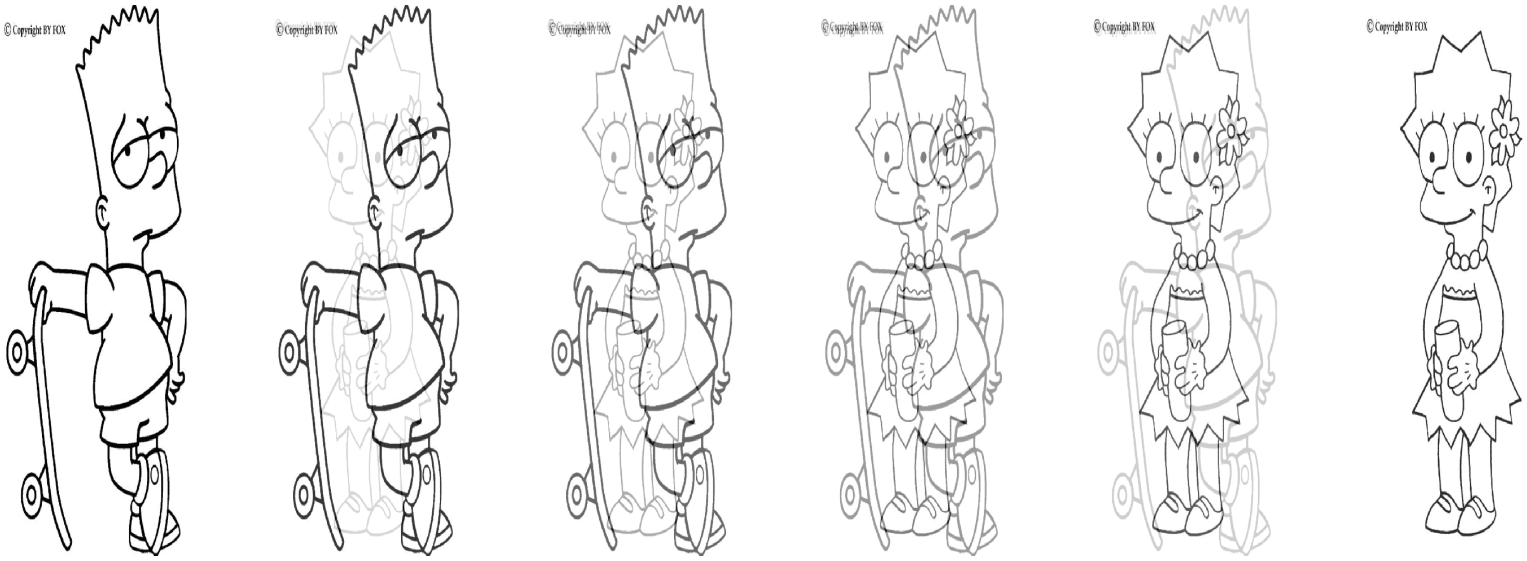
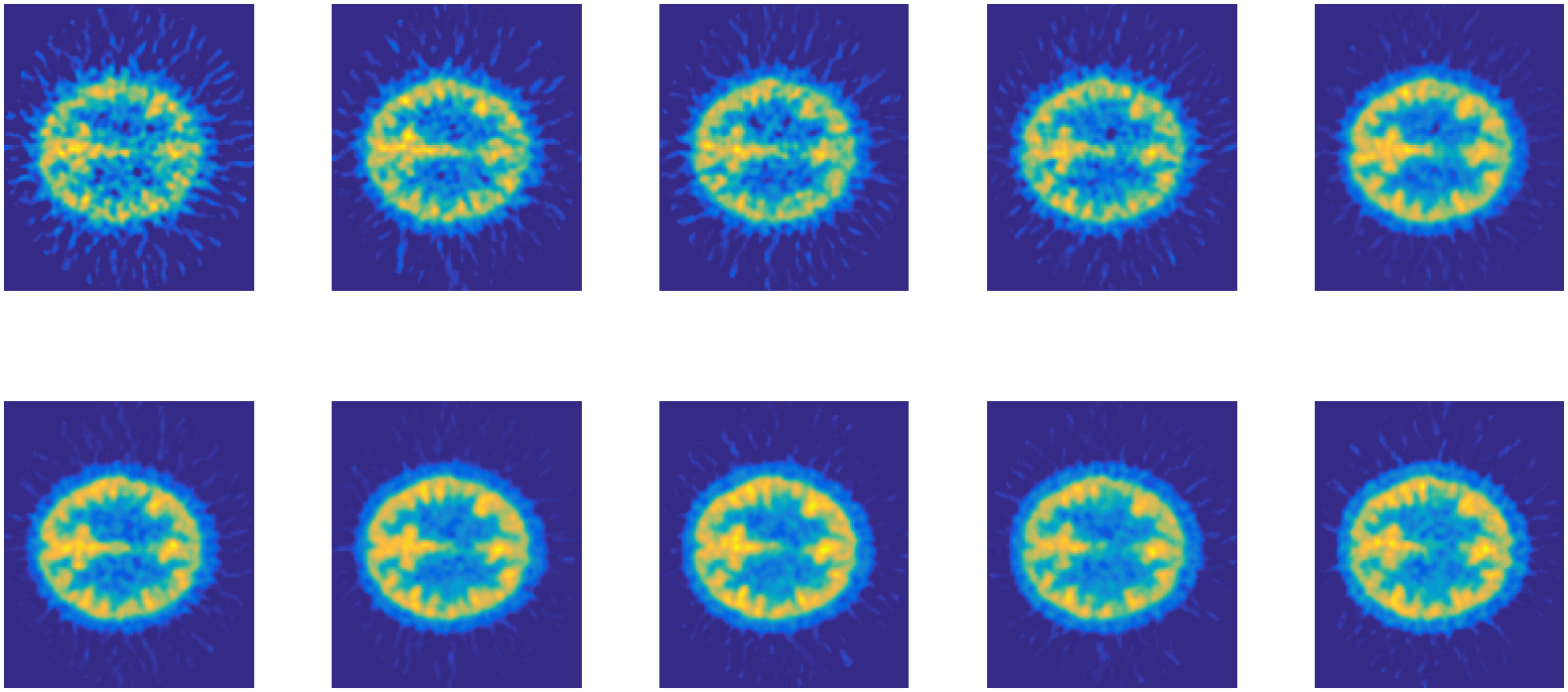


- 2 sources
- weights:

Bart	1	0.8	0.6	0.4	0.2	0
Lisa	0	0.2	0.4	0.6	0.8	1

Medical imaging



- number of sources?
- source images
- time activity of the source

Mathematical description

Linear model of $p \times n$ matrix

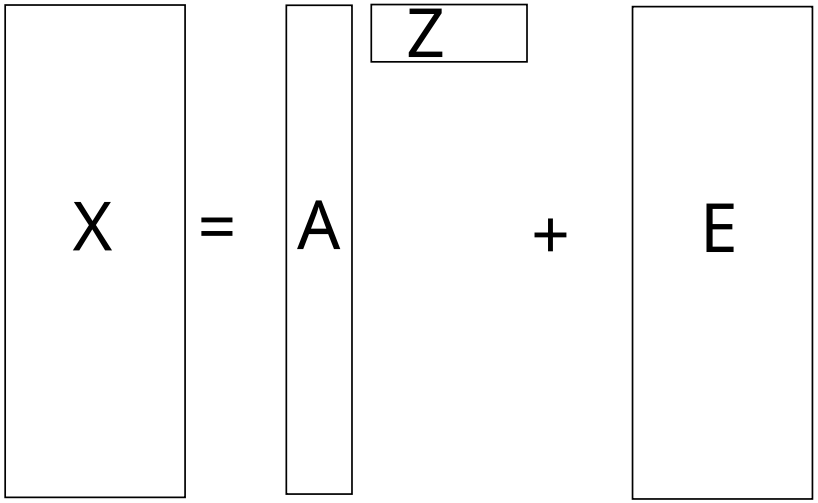
$$d_i = Ax_i + e_i, i = 1..n \qquad\qquad D = AX^T + E,$$

where

A is the $p \times r$ matrix of source images, $r < \min(n, p)$

X is the $n \times r$ matrix of time activities,

E is the $p \times n$ noise matrix (Gaussian)



The case of known X: least squares

Consider decomposition $D = AX^\top$

$$\underline{d}_j = \underline{a}_j X^\top \quad \Longrightarrow \quad \underline{d}_j^\top = X \underline{a}_j^\top,$$

for known D, X the estimate of A is

$$\hat{\underline{a}}_j^\top = (X^\top X)^{-1} X^\top \underline{d}_j^\top, \quad \Sigma_{a_j} = (X^\top X)^{-1},$$

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In matrices

$$\hat{A} = DX(X^T X)^{-1}, \quad p(A) = \mathcal{N}(\hat{A}, I, \Sigma_{a_j}) = \prod_j \mathcal{N}(\hat{\underline{a}}_j, \Sigma_{a_j}),$$

where $\mathcal{N}(\hat{A}, I, \Sigma_{a_j})$, is *matrix normal distribution*, with row and column covariance matrices.
Useful expectation

$$\mathrm{E}(A^T A) = \sum_{j=1}^{p} \underline{a}_j \underline{a}_j^T = \sum_{j=1}^{p} \left(\hat{\underline{a}}_j \hat{\underline{a}}_j^T + \Sigma_{a_j} \right) = \hat{A}^T \hat{A} + p \Sigma_{a_j}$$

Matrix Normal distribution

Matrix normal distribution,

$$\mathcal{N}(\hat{A}, \Sigma_{row}, \Sigma_{col}) \propto |\Sigma_{row}|^{-\frac{r}{2}} |\Sigma_{col}|^{-\frac{p}{2}} \exp\left(-\frac{1}{2}\mathrm{tr}\left((A-\hat{A})\Sigma_{col}^{-1}(A-\hat{A})\Sigma_{row}^{-1}\right)\right),$$

with row and column covariance matrices. It holds that

$$p(\mathrm{vec}(A)) = \mathcal{N}(\mathrm{vec}(\hat{A}), \Sigma_{col} \otimes \Sigma_{row}).$$

using Kronecker product $A \otimes B = \begin{bmatrix} a_{11}B & \cdots & a_{1n}B \\ \vdots & \ddots & \vdots \end{bmatrix}$.

Useful results

$$\mathrm{E}(A^T ZA) = \hat{A}^T ZA + \mathrm{tr}(\Sigma_{row} Z)\Sigma_{col}$$

Source Separation

Model

$$D = AX^T + E,$$

Both images A and time activities X are **unknown**!

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Issue: ambiguity

$$\begin{aligned} AX^T = \quad & ATT^{-1}X^T \\ & (AT)(T^{-1}X^T) \quad = \overline{AX}^T \end{aligned}$$

matrix T includes both scaling and rotation.

Solution: additional constraints.

Principal component analysis

Consider n, p-dimensional vectors $d_i, i = 1, \ldots n$, and their covariance matrix

$$S = \frac{1}{n} \sum (d_i - \overline{d})(d_i - \overline{d})^T.$$

Then r-dimensional vectors x_i,

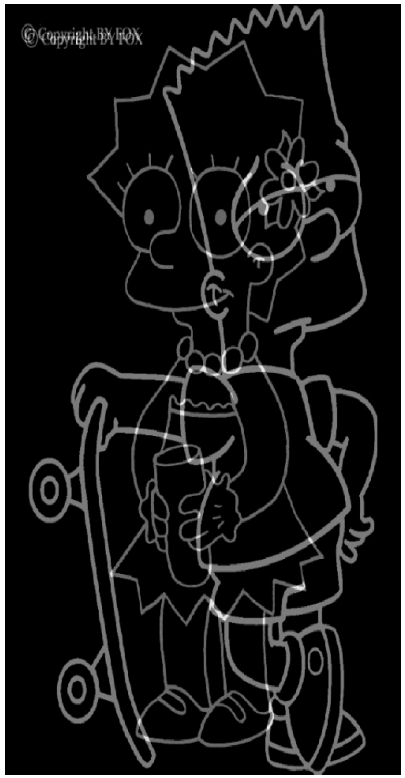
$$x_i = U(:, 1:r)^T d_i, \qquad S = U \Lambda U^T,$$

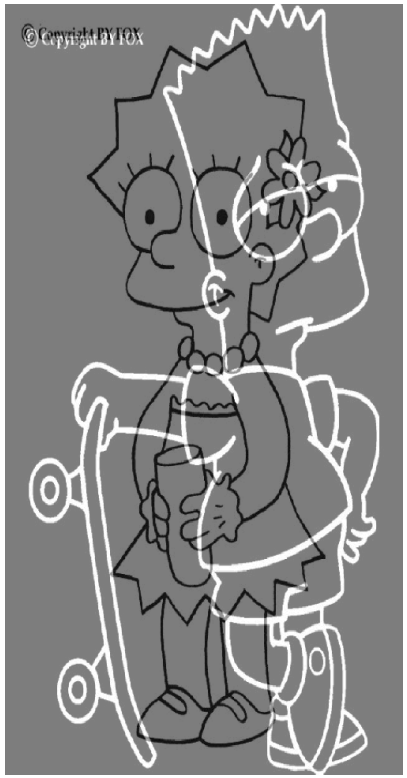
has maximum variance from all possible projections to r dimensions. U are eigenvectors of S sorted with decreasing eigenvalue.

Matrix SVD approach:

$$\begin{aligned} D &\overset{\text{svd}}{=} USV, \\ &= \underbrace{(US)}_{\hat{A}} \underbrace{(V)}_{\hat{X}^T} \end{aligned}$$

Application to sequences of images





- Popular few years ago for speed of computation,
- Requires to find the rotation matrix T,
- Can we do better?
  - Independent Component Analysis (higher order moments)?
  - Structural priors
  - Non-negative matrix factorization

Probabilistic PCA [Tipping, Bishop, 1999]

Consider model:

$$p(d_i|A, x_i, \sigma) = \mathcal{N}(Ax_i, \sigma I_p), \qquad p(x_i) = \mathcal{N}(0, I_r),$$

Marginalization of $p(d_i, x_i|A, \sigma)$ over x_i (lesson 1) yields

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Marginalization of $p(d_i, x_i|A, \sigma)$ over x_i (lesson 1) yields

$$p(d_i|A, \sigma) = \mathcal{N}(0, C), \qquad p(D|A, \sigma) = \prod_i \mathcal{N}(0, C)$$

$$C = AA^T + \sigma I_p,$$

The likelihood is

$$p(D|A, \sigma) \propto \prod_i |C|^{-\frac{1}{2}} \exp\left(-\frac{1}{2} d_i^\top C^{-1} d_i\right) \propto |C|^{-\frac{n}{2}} \exp\left(-\frac{1}{2}\mathrm{tr}(C^{-1} \sum_i d_i d_i^\top)\right)$$

$$\propto |C|^{-\frac{n}{2}} \exp\left(-\frac{1}{2}\mathrm{tr}(C^{-1} DD^\top)\right)$$

Maximum Likelihood

Maximum likelihood

$$p(D|A,\sigma) \propto \left|AA^T + \sigma I_p\right|^{-\frac{n}{2}} \exp\left(-\frac{1}{2}\mathrm{tr}((AA^T + \sigma I_p)^{-1} DD^\top)\right)$$

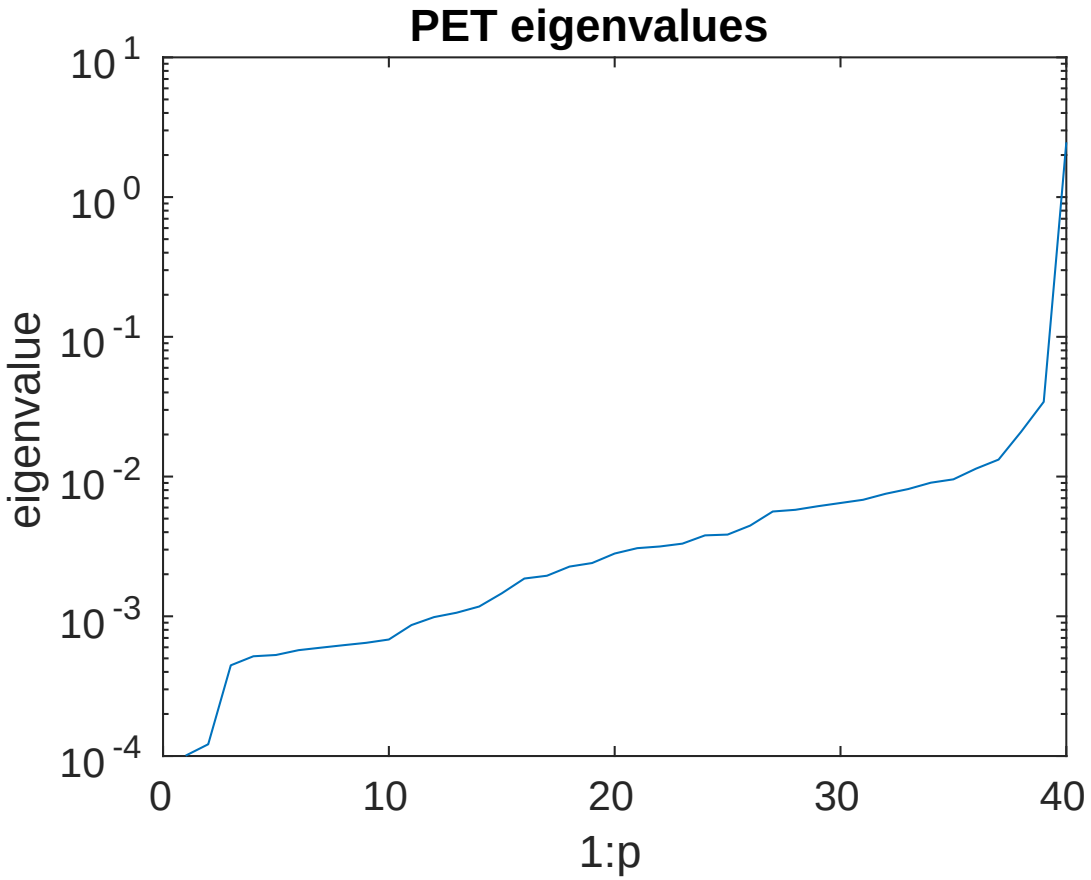
for $\hat{A}$and $\hat{\sigma}$ for given r:

$$\hat{A} = U_{1:r}(\Lambda_{1:r} - \hat{\sigma} I)^{\frac{1}{2}} T, \qquad \hat{\sigma} = \frac{1}{d-r} \sum_{i=r+1}^{n} \lambda_i,$$

where $S = U\Lambda U^T$ is eigen-decomposition of $S = \frac{1}{n}\sum d_i d_i^T$.

- distinction from PCA: subtraction of the noise

Image Sequence (PET)



- no plateau

Alternative: EM algorithm

Maximize

$$p(X|D) \propto \int p(D|A,X)p(A)dA$$

where

$$p(A) = \mathcal{N}(0, I, I) \propto \exp\left(-\frac{1}{2}\text{tr}\ (AI_rA^TI_p)\right),$$

Joint model

$$\begin{aligned}\log p(A, D|X) = &-\frac{1}{2}\sigma^{-1}\text{tr}(D - AX^T)^T(D - AX^T) \\ &- \frac{1}{2}\text{tr}\ (AI_rA^TI_p)\end{aligned}$$

Conditional $p(A|D, X, \sigma)$ for known X and σ

$$\begin{aligned}p(A|X, D, \sigma) &= \mathcal{N}(\hat{A}, I, \Sigma_A) \\ \hat{A} &= DX(X^TX + \sigma I)^{-1}, \qquad \Sigma_A = (\sigma^{-1}X^TX + I)^{-1}\end{aligned}$$

The EM algorithm

Standard form for

E-step (over A): $q(X|X^{(j)}) = \int \log p(D, A|X) p(A|X^{(j)}, D) dA$

M-step (of X): $X^{(j+1)} = \arg\max_\theta q(X|X^{(j)})$

The $q(X|X^{(j)})$ factor is:

$$\begin{aligned}
\log p(A, X, D) &= -\frac{1}{2}\sigma^{-1}\mathrm{tr}(D - AX^T)^T(D - AX^T) \; - \frac{1}{2}\mathrm{tr}\,(AI_rA^TI_p) \\
&= -\frac{1}{2}\sigma^{-1}\mathrm{tr}\left(D^TD - XA^TD - DAX^T + XA^TAX^T + A^TA\sigma\right) \\
q(X|X^{(j)}) &= \mathrm{E}_A\left\{-\frac{1}{2}\sigma^{-1}\mathrm{tr}\left(D^TD - XA^TD - DAX^T + XA^TAX^T\right)\right\} \\
&= -\frac{1}{2}\sigma^{-1}\mathrm{tr}\left(D^TD - X\hat{A}^TD - D^T\hat{A}X^T + X(\hat{A}^T\hat{A} + p\Sigma_A)X^T\right) \\
&\propto \mathcal{N}(\hat{X}, \Sigma_X), \\
&\hat{X} = D^T\hat{A}(\hat{A}^T\hat{A} + p\Sigma_A)^{-1},
\end{aligned}$$

Summary of the EM Algorithm

Iterate two least squares problems:

1. $\hat{A} = DX(X^T X + \sigma I)^{-1}, \Sigma_A = (\sigma^{-1} X^T X + I)^{-1}$
2. $\hat{X} = D^T \hat{A}(\hat{A}^T \hat{A} + p\Sigma_A)^{-1}$

If you initialize at orthogonal solution $\hat{A} = U_{1:r}\Lambda_{1:r}$ it will be operating only on λs.

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- Why I do not care about variance of the prior, λI?
  - ambiguity $AX^T = ATT^{-1}X^T$
- We can incorporate many more assumptions!
  - positivity
  - $p(A) = \mathcal{N}(0, I, \text{diag}(\alpha))$

Positivity constraint on X

What if we impose prior $p(X_{i,j}) = \begin{cases} 1 & \text{if } X_{i,j} > 0 \\ 0 & \text{otherwise.} \end{cases}$

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M-step (of X): $X^{(j+1)} = \arg\max_\theta q(X|X^{(j)})$ s.t. $X_{i,j} > 0, \forall i, j$

Recall the q function:

$$q(X|X^{(j)}) = \mathcal{N}(\hat{X}, \Sigma_X), \;\; \hat{X} = D^T \hat{A}(\hat{A}^T \hat{A} + p\Sigma_A)^{-1},$$

With additional constraint we obtain truncated Normal distribution

Extreme $\hat{\hat{X}}$is $\hat{\hat{X}}_{i,j} = \begin{cases} \hat{X}_{i,j} & \text{if } \hat{X}_{i,j} > 0 \\ 0 & \text{otherwise} \end{cases}$

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What if we impose prior $p(A_{i,j}) = t\mathcal{N}(0, I, I, [0, \infty])$

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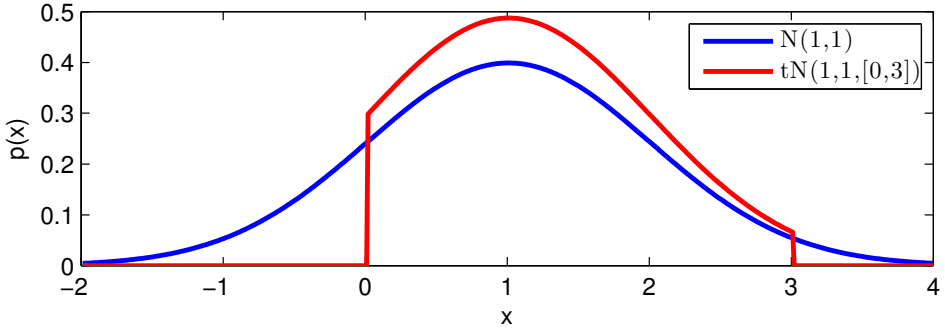
$$q(X|X^{(j)}) = \mathrm{E}_A \left\{ -\frac{1}{2}\sigma^{-1}\mathrm{tr}\left(D^T D - XA^T D - DAX^T + XA^T AX^T\right) \right\}$$



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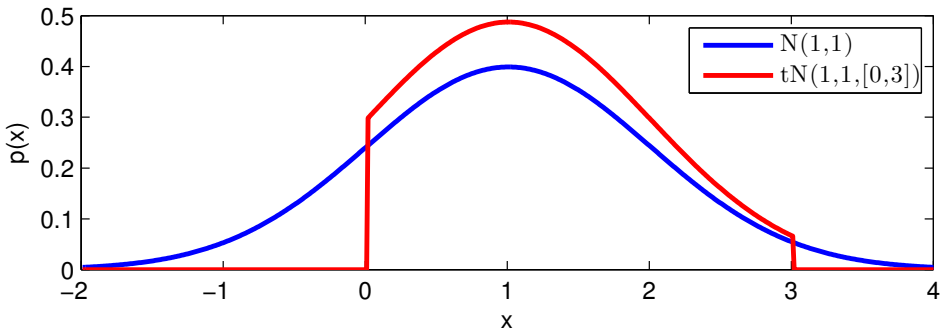

$\mathrm{E}(A) \approx \tilde{A}_{i,j} = \mathcal{N}(\hat{A}, I, \mathrm{diag}(\Sigma_A))$ product of univariate tN. Moments of $t\mathcal{N}(\hat{a}, \sigma_a)$ are

$$\mathrm{E}(a) = \hat{a} + \frac{\phi(\alpha) - \phi(\beta)}{Z_a},$$

$$\mathrm{E}(a^2) = \sigma_a^2 \left\{ 1 + \frac{\alpha\phi(\alpha) - \beta\phi(\beta)}{Z_a} \right\}$$

where $\alpha = \frac{-\hat{a}}{\sqrt{\sigma_a}}$, $\beta = \infty$, $\phi(x) = \frac{1}{\sqrt{2\pi}} \exp(-x^2/s)$, $Z = \Phi(\beta) - \Phi(\alpha)$, $\Phi(x) = \frac{1}{2}(1 + \mathrm{erf}(x))$.

Variational Bayes for PCA

Bayes rule

$$p(A, X|D) \propto p(D|A, X)p(A)p(X)$$

where

$$p(A) = \mathcal{N}(0, I, I) \propto \exp\left(-\frac{1}{2}\text{tr}\ (AI_rA^TI_p)\right),$$

$$p(X) = \mathcal{N}(0, I, I) \propto \exp\left(-\frac{1}{2}\omega_X\text{tr}(XI_rX^TI_n)\right)$$

we can no longer have both variances one.

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we can no longer have both variances one.

$$q(A, X) \approx q(A)q(X)$$

Yields:

1. $\hat{A} = D\hat{X}(\hat{X}^T\hat{X} + n\Sigma_X + I)^{-1}, \Sigma_A = (X^TX + n\Sigma_X + I)^{-1}$
2. $\hat{X} = D^T\hat{A}(\hat{A}^T\hat{A} + p\Sigma_A + \omega_X I)^{-1}, \Sigma_X = (\hat{A}^T\hat{A} + p\Sigma_A + \omega_X I)^{-1},$

In practice we need to estimate precision of data ω_D, ω_X needing expectations

$$\mathrm{E}_{A,X}\left(XA^TAX^T\right)$$

Toy matrix decomposition

Consider 1×1 matrix d, decomposed

$$p(d|a, x) = \mathcal{N}(ax, r_e),$$

Find a, x.

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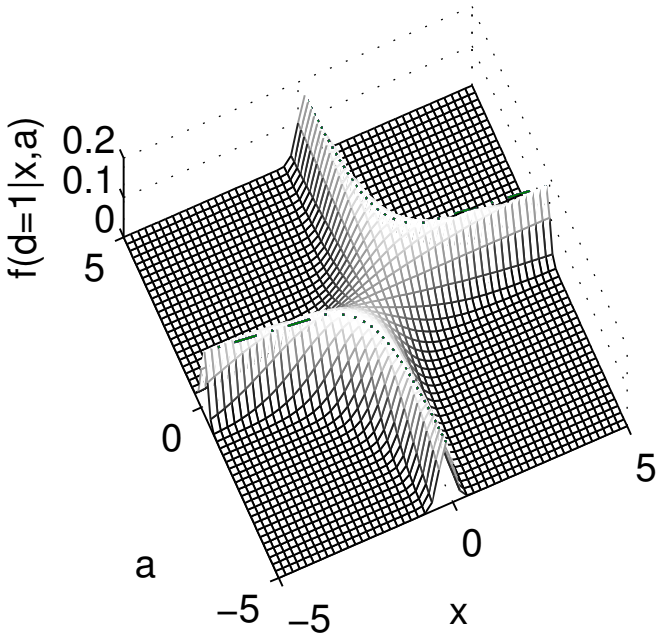
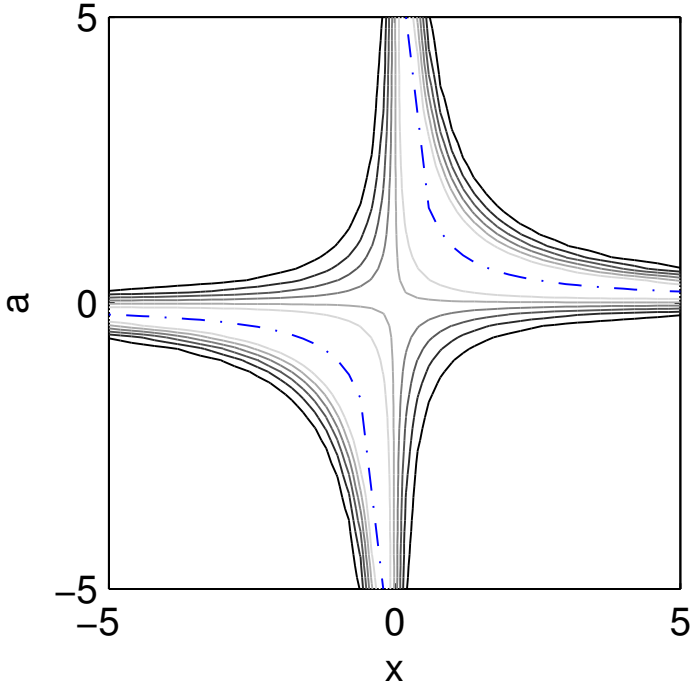
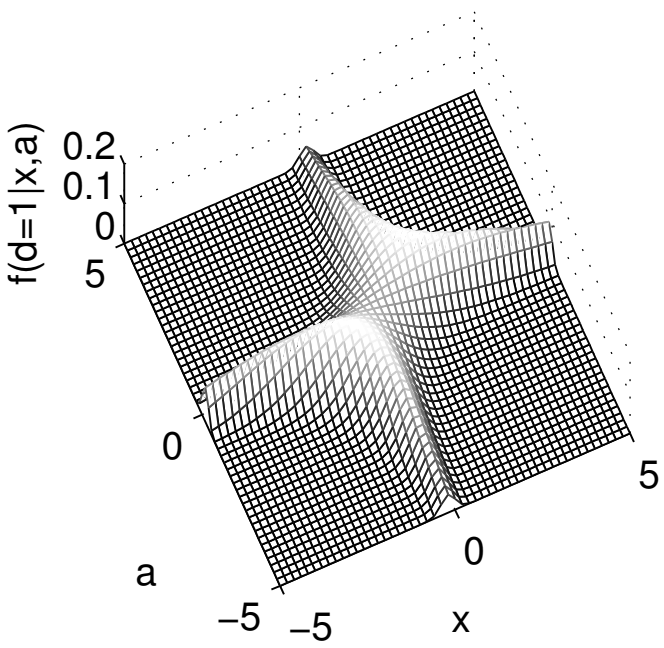


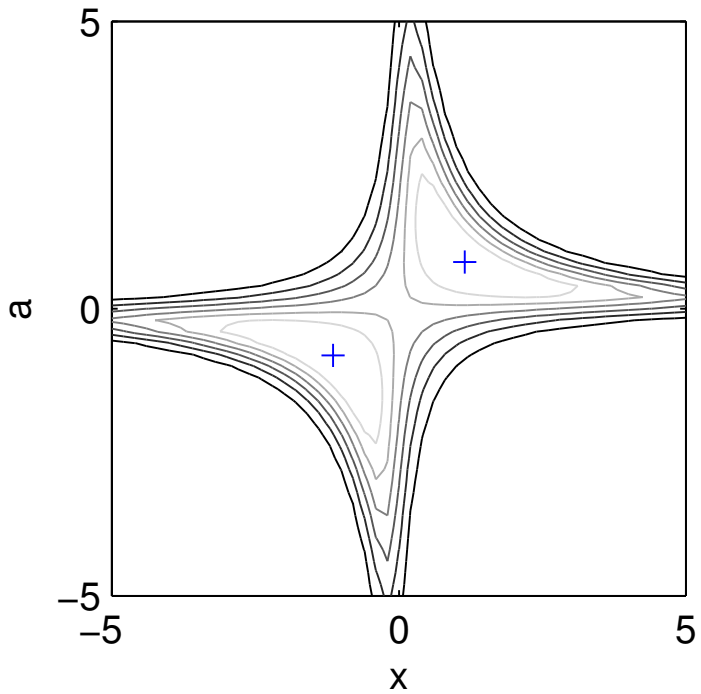


Toy matrix decomposition

Consider 1×1 matrix d, decomposed

$$p(d|a,x) = \mathcal{N}(ax, r_e),$$
$$p(x) = \mathcal{N}(0, r_x),$$
$$p(a) = \mathcal{N}(0, r_a)$$





Toy maximum likelihood

Joint distribution:

$$\log p(d, a, x) \propto -\frac{1}{2r_e}(d - ax)^2 - \frac{1}{2r_a}a^2 - \frac{1}{2r_x}x^2$$

Find

$$\hat{x}, \hat{a} = \arg\max_{a,x}(\log p(d, a, x))$$

For $d < \frac{r_e}{\sqrt{r_a r_x}}$,

$$\hat{x} = 0, \hat{a} = 0,$$

For $d \geq \frac{r_e}{\sqrt{r_a r_x}}$,

$$\hat{x} = \pm\left(d\sqrt{\frac{r_x}{r_a}} - \frac{r_e}{r_a}\right)^{\frac{1}{2}}, \qquad \hat{a} = \pm\left(d\sqrt{\frac{r_a}{r_x}} - \frac{r_e}{r_x}\right)^{\frac{1}{2}}.$$

Note that the product of the maxima is

$$\hat{a}\hat{x} = d - \frac{r_e}{\sqrt{r_a r_x}}.$$

Marginal likelihood (PPCA)

Joint distribution:

$$\log p(d, a, x) \propto -\frac{1}{2r_e}(d - ax)^2 - \frac{1}{2r_a}a^2 - \frac{1}{2r_x}x^2$$

Marginal

$$\begin{aligned} p(a|d) &\propto \int p(d, a, x)dx \\ &\propto \exp(-\frac{1}{2}d^2(a^2 r_x + r_e)^{-1})\sqrt{r_e r_x}(a^2 r_x + r_e)^{-\frac{1}{2}}, \end{aligned}$$

with maximum:

$$\hat{a} = \begin{cases} \frac{\sqrt{d^2 - r_e}}{\sqrt{r_x}} & \text{if } d^2 > r_e, \\ 0 & \text{otherwise}, \end{cases}$$

Variational Bayes

Joint distribution:

$$\log p(d, a, x) \propto -\frac{1}{2r_e}(d - ax)^2 - \frac{1}{2r_a}a^2 - \frac{1}{2r_x}x^2$$

Factor $q(a|d)$

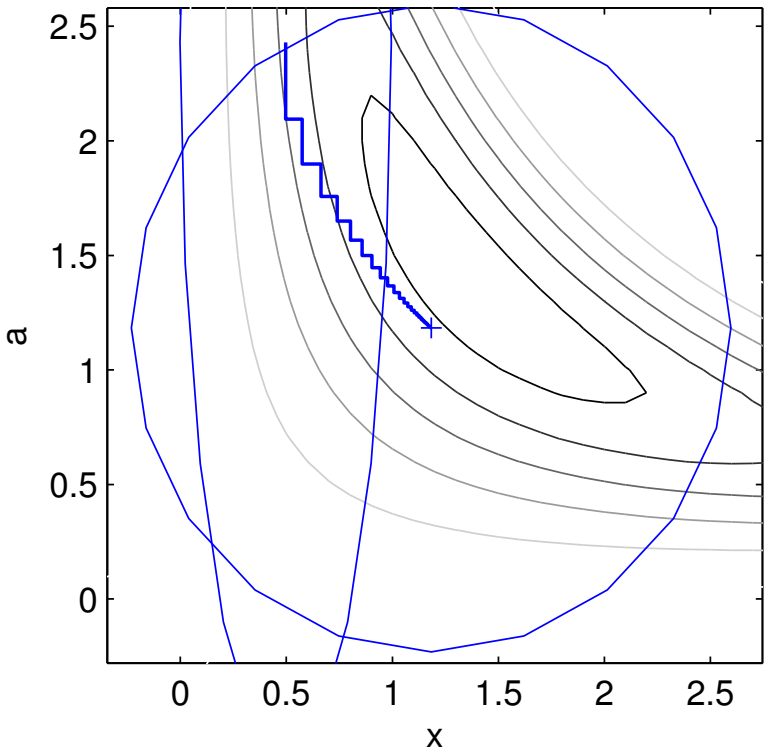
$$\begin{aligned} p(a|d) &\propto \exp\left(\mathrm{E}_x\left[-\frac{1}{2r_e}(d - ax)^2 - \frac{1}{2r_a}a^2\right]\right) \\ &= \mathcal{N}(\hat{a}, \sigma_a), \\ \hat{a} &= \sigma_a d \langle x \rangle r_e^{-1}, \quad \sigma_a^{-1} = \langle a^2 \rangle r_e^{-1} + r_a^{-1} \end{aligned}$$

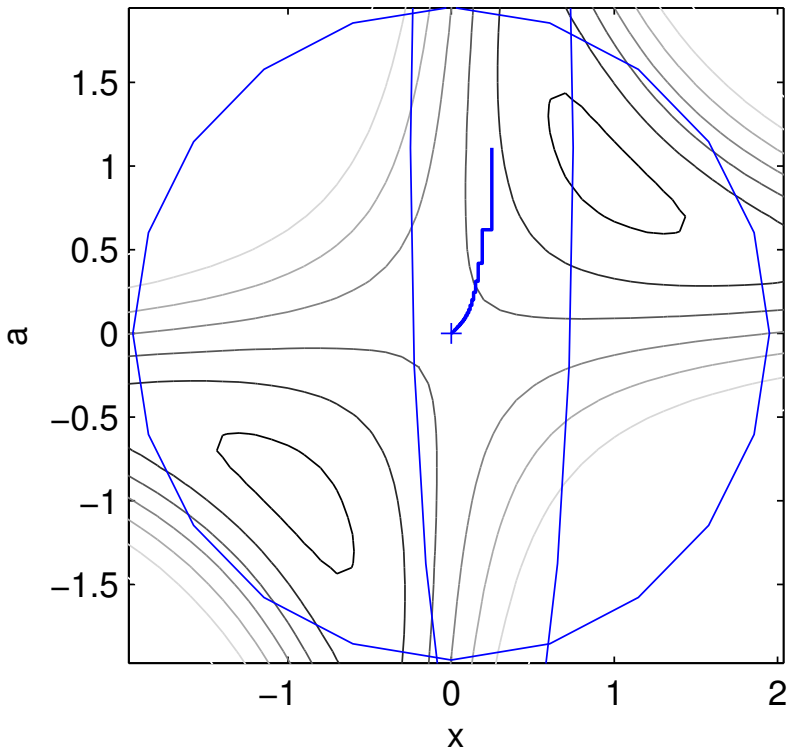
Factor $q(x|d)$

$$\begin{aligned} p(a|d) &\propto \exp\left(\mathrm{E}_x\left[-\frac{1}{2r_e}(d - ax)^2 - \frac{1}{2r_a}a^2\right]\right) \\ &= \mathcal{N}(\hat{a}, \sigma_a), \\ \hat{a} &= \sigma_a d \langle x \rangle r_e^{-1}, \quad \sigma_a^{-1} = \langle a^2 \rangle r_e^{-1} + r_a^{-1} \end{aligned}$$

Convergence of VB

1. compute $\hat{a}, \sigma_a$, and $\langle a \rangle = \hat{a}$, $\langle a^2 \rangle = \hat{a}^2 + \sigma_a$,
2. compute $\hat{x}, \sigma_x$, and $\langle x \rangle = \hat{x}$, $\langle x^2 \rangle = \hat{x}^2 + \sigma_x$,





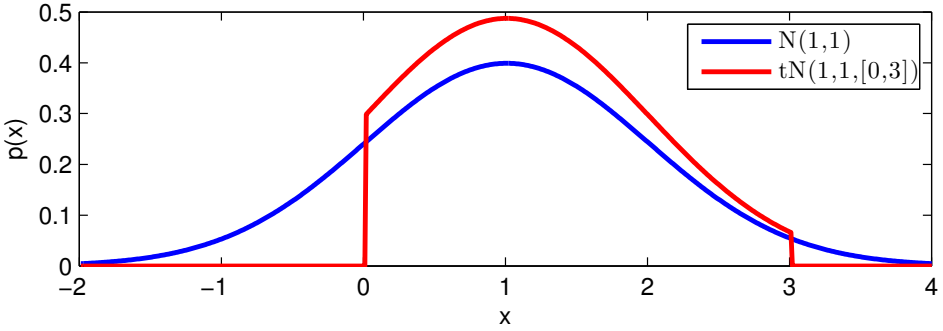
Positive support

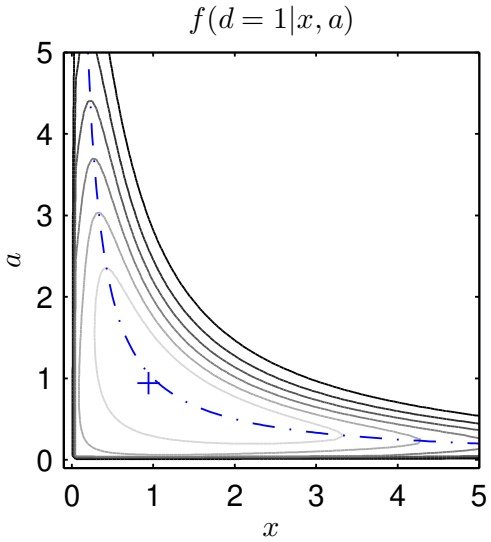
What if we are interested only in the positive solution?

Positive support

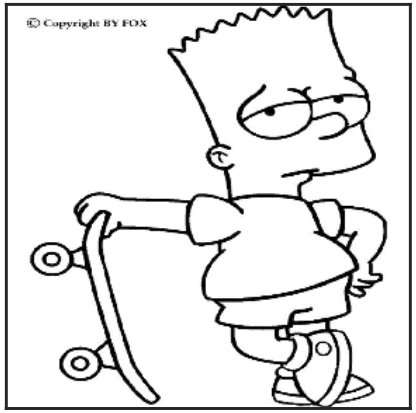
What if we are interested only in the positive solution?

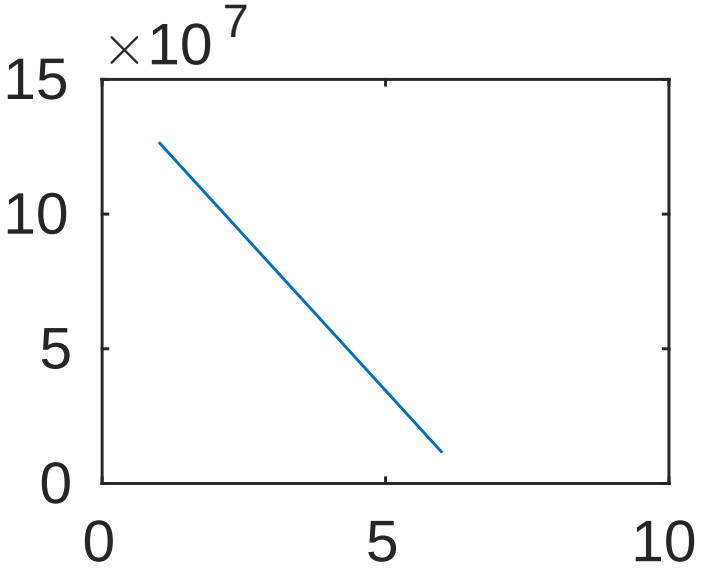
$$
\begin{aligned}
p(d|a,x) &= \mathcal{N}(ax, r_e), \\
p(x) &= t\mathcal{N}(0, r_x, \langle 0, \infty \rangle) \propto \mathcal{N}(0, r_x)\chi(x > 0), \\
p(a) &= t\mathcal{N}(0, r_a, \langle 0, \infty \rangle) \propto \mathcal{N}(0, r_a)\chi(a > 0),
\end{aligned}
$$

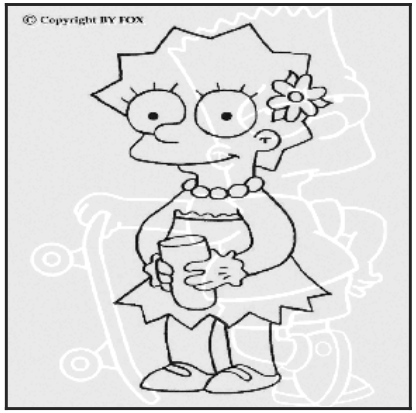


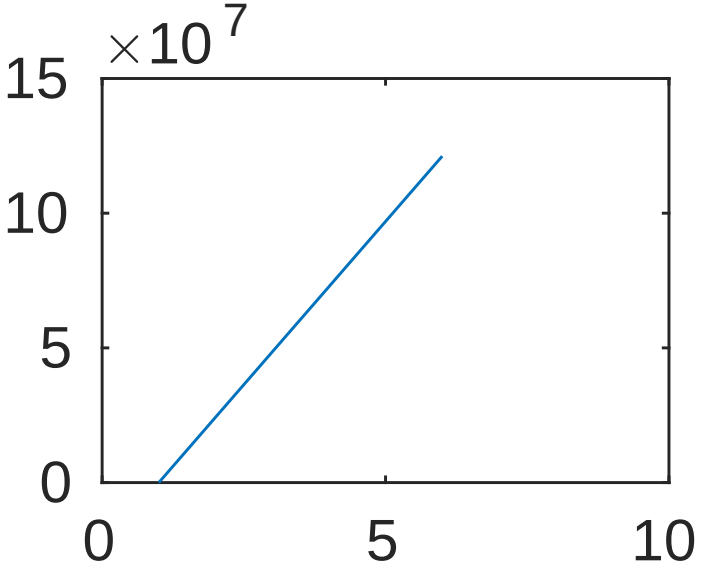


Non-negative Matrix Factorization (NMF)









Extensions: Multivariate PCA

1. Automatic Relevance Determination (#of factors):

$$p(A) = \mathcal{N}(0, I, \text{diag}(\alpha)),$$

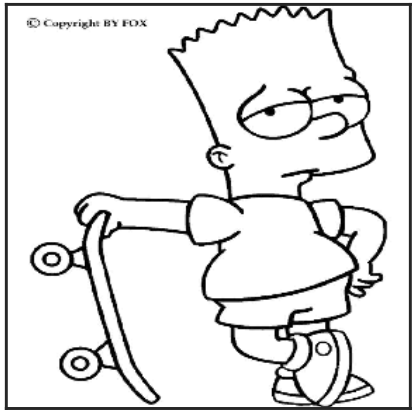
2. Automatic Relevance Determination (#pixels):

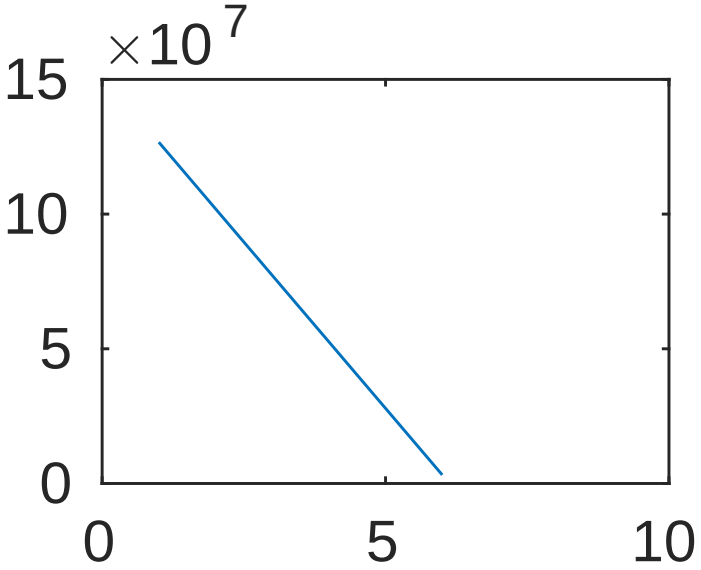
$$p(\text{vec}(A)) = \mathcal{N}(0, \text{diag}(\alpha)),$$

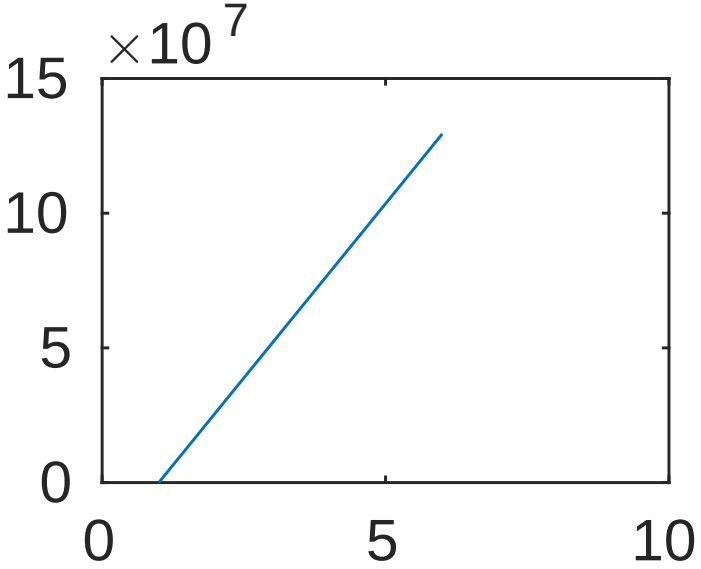
   where α

3. many more

Sparse Non-negative Matrix Factorization (SNMF)





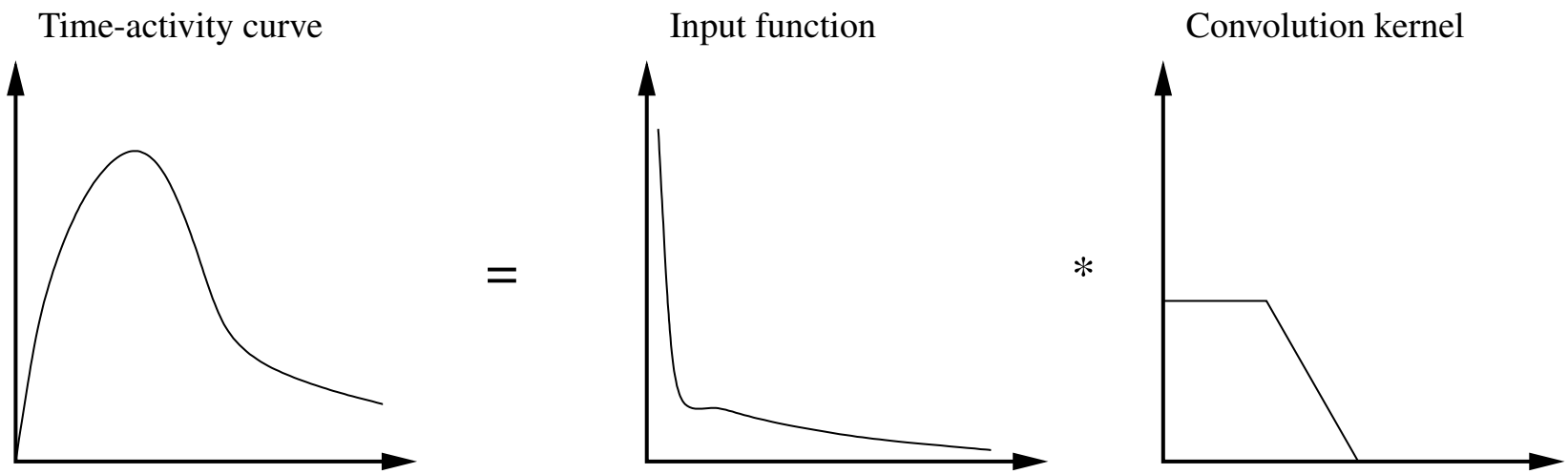


Assignment

Matrix factorization		points
EM or VB, Bart-Lisa		8
+ positive support		+2

BSS with deconvolution

Time-activity curves for brain imaging are results of convolution.



Model of the curve

$$x = b * w = Bw, \qquad B = \begin{pmatrix} b_1 & 0 & 0 & 0 \\ b_2 & b_1 & 0 & 0 \\ \dots & b_2 & b_1 & 0 \\ b_n & \dots & b_2 & b_1 \end{pmatrix}$$

Unknows are b and w. Kernel is sparse.

BSS with deconvolution

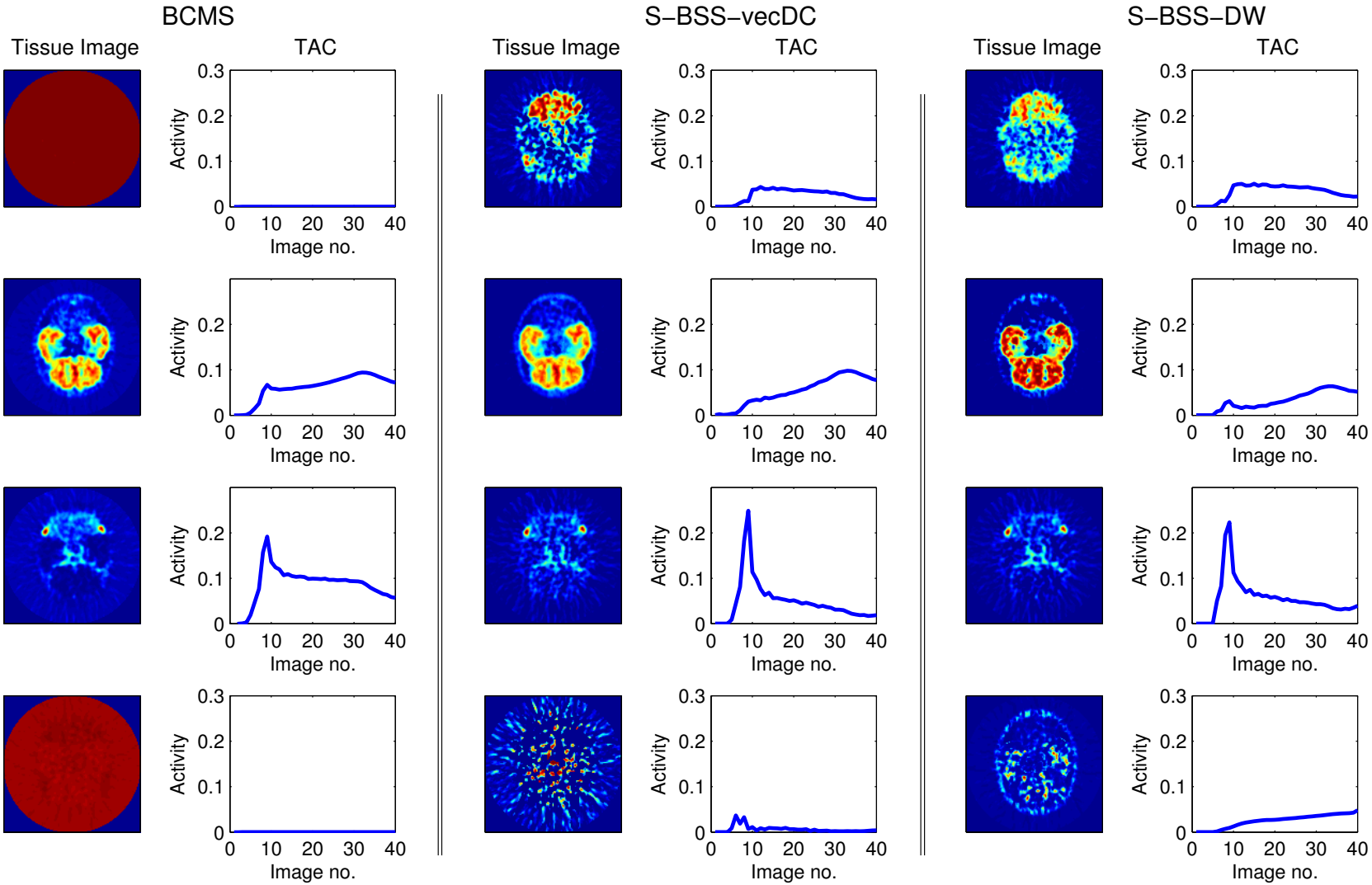


Image deconvolution

(a) out-of-focus blur

Image deconvolution

Mathematical model:

$$d = Ax + e$$

where

d	is the observed (blured) image,	
x	is the original (true) image,	unknown
A	is the convolution matrix,	
e	is the measurement (model) error	

Find: x, A, $var(e)$

Image deconvolution

Mathematical model:

$$d = Ax + e$$

where

d	is the observed (blured) image,	
x	is the original (true) image,	unknown
A	is the convolution matrix,	unknown kernel
e	is the measurement (model) error	

Find: x, A, $var(e)$

Image deconvolution

Mathematical model:

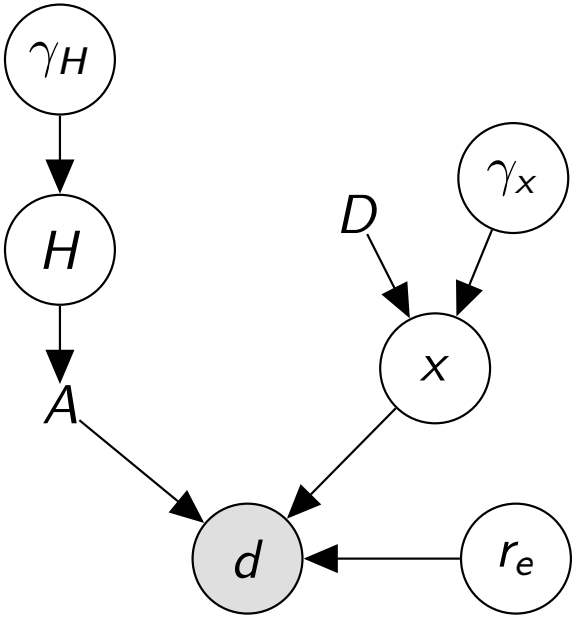
$$d = Ax + e$$

where

d	is the observed (blured) image,	
x	is the original (true) image,	unknown
A	is the convolution matrix,	unknown kernel
e	is the measurement (model) error	unknown variance

Find: x, A, $var(e)$

Model



- number of unknowns $> 3\times$ higher than number of observations
- ARD coefficients γ_H, γ_x, r_e,
- approximations of the covariance matrices by diagonal

Results:

(a) out-of-focus blur

(b) blind deconvolution